HEALTHBRIDGE PROJECT MANAGEMENT PLAN

UCSF Health Patient Navigator (RAG Chatbot) • Residency Day Execution

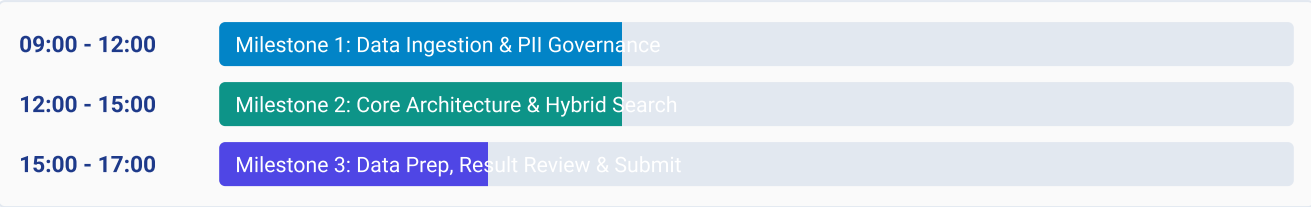
Technical PM: Shabbar Al
Team Size: 7 Members
Duration: 8 Hours

PURE GOVERNANCE
100% technical task delegation across specialized engineers; PM focuses strictly on timeline, workflow, and blockers.

REVIEW GATEKEEPER
PM operates key code-review checkpoints and enforces mandatory review pairings (e.g., evaluating Ravalika's extraction PRs).

REPORTING OWNER
Direct ownership of academic compliance, authoring the `01_ingestion` report and repository `README`.

1. HIGH-LEVEL 8-HOUR EXECUTION SCHEDULE



2. DELEGATION & GOVERNANCE MATRIX (MILESTONES 1 & 2)

Milestone & Time	Team Members	Detailed Member Task Breakdown	PM Oversight & Review Actions
Milestone 1 09:00 - 12:00 *Ingestion & PII*	Zaineb Syeda Ravalika Gundu	**Zaineb:** Executes Stage 1 crawler respecting `robots.txt` limits (≤1 rps) and sitemap seeds. **Ravalika:** Handles Stage 2 extraction, pulling HTML content via `trafilatura` and tables/text from PDFs via `pdfplumber`.	• Set up GitHub project board, issue cards, and main branch protections. PM GATE 1 Review and approve Ravalika's extraction PR; audit raw manifests and PII logs.
Milestone 2 12:00 - 15:00 *Architecture & Search*	Abul Hasan Harsha Vardhan Sai Chandra Manish Karki	**Abul:** Directs system architecture & implements Stage 3 structure-aware chunking (`sa-v1`). **Harsha:** Builds Stage 4 `bge-base-en-v1.5` embeddings & ChromaDB vector store. **Sai:** Implements Stage 7 BM25 search & Reciprocal Rank Fusion. **Manish:** Configures Stage 8 cross-encoder reranking.	• Monitor pipeline transitions from chunking to ChromaDB and BM25 index generation. PM GATE 2 Audit index health cards (`index_card.json`) and verify B2 baseline connectivity.

3. MILESTONE 3 EXECUTION MATRIX (15:00 - 17:00)

Focus Area	Team Members	Detailed Member Task Breakdown	PM Oversight & Review Actions
Data Prep, Evaluation & Lock 15:00 - 17:00	Ravalika Gundu Manish Karki Shabbar AI (PM) Abul Hasan Zaineb Syeda	**Data Scientists (Ravalika & Manish):** Manually prepare, verify, and cross-check the golden set (`eval/golden_set.jsonl`) against live UCSF source pages. **PM & Architect (Shabbar & Hasan):** Independently review chatbot outputs against real answers to verify model accuracy, grounding, and citation validity. **Zaineb:** Finalizes Streamlit Chat UI & safety guardrails.	• Oversee Streamlit UI and safety guardrail testing. PM GATE 3 Finalize `01_ingestion` report and project README, verify all PR approvals, and lock the repo for submission.

4. TPM ACCURACY & VERIFICATION PROTOCOL

Ground-Truth Preparation (Data Scientists)

- Manual verification of live UCSF Health pages across all 6 golden set categories.
- Double-human verification protocol (author + peer reviewer sign-off).
- Ensuring exact 25-word evidence quote limit for support testing.

Accuracy Audit (PM & Solutions Architect)

- Direct output review comparing model answers against gold ground-truth.
- Evaluation of refusal correctness on unanswerable and ambiguous queries.
- Audit of citation support overlap (≥0.35 threshold) and quote truncation.

5. SUMMARY OF TEAM ROLE ALLOCATIONS

Project Management & Architecture: Shabbar AI (PM) & Abul Hasan (Solutions Architect) handle governance, baseline reviews, and model output accuracy audits.

Data Science & Curation: Ravalika Gundu & Manish Karki manage extraction, golden set manual verification, reranking, and evaluation harnesses.

Backend & Frontend Engineering: Harsha Vardhan (Embeddings/Vector DB), Sai Chandra (BM25/Hybrid), and Zaineb Syeda (Scrape/Generation/UI/Safeguards) handle core software implementation.